

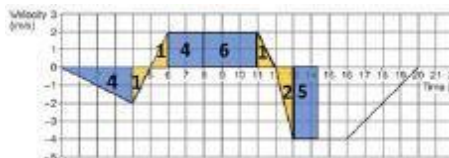
Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Draws first stage of graph correctly.	AO3.3	B1	Line connecting (0, 0) to (4, 3).
	Draws second stage correctly Line moving down by 6 m s ⁻¹ in 5 s	AO3.3	B1F	Line connecting (4, 3) to (9, -3).
	Deduces final stage correctly Horizontal line of length 33/ v where v is the speed of the vehicle at the end of the second stage.	AO2.2a	B1F	Line connecting (9, -3) to (20, -3).
(b)	Finds at least one non-rectangular area. As typical solution Or t between 0 & 6.5: Area of triangle $= 3 \times 6.5 \div 2 = 9.75$ m (forward) t between 6.5 & 20: Area of trapezium $= 0.5(13.5 + 11) \times (-3) = -36.75$ m (backwards)	AO1.1a	M1	t between 0 & 4: Area of triangle $= 3 \times 4 \div 2 = 6$ m (forward) t between 4 & 6.5: Area of triangle $= 3 \times 2.5 \div 2 = 3.75$ m (forward) t between 6.5 & 9: Area of triangle $= 2.5 \times -3 \div 2 = -3.75$ m (backwards) t between 9 and 20: Area of rect. $= -3 \times 11 = -33$ m (backwards)
	Subtracts <i>their</i> areas below axis from areas above axis. OE	AO1.1a	M1	Displacement from O is $9.75 - 36.75 = -27$ m
	Calculates distance from P correctly. CAO. Units not required	AO3.2a	A1	Distance is 27 m
				Total 6 marks

Q2.

Marking Instructions	AO	Marks	Typical Solution
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(a)	Finds the steepest gradient. Ignore units. Do not allow -4.	AO1.1b	B1	4 m s ⁻²	
(b)	Shows evidence of determining areas above and below the time axis for values of t between 7 and 9 inclusive Evidence may include values or indication on a diagram	AO3.1b	M1		
	Deduces t_1 value correctly.	AO2.2a	A1		$t_1 = 8$
	Shows evidence of determining areas above and below the time axis for values of t between 13 and 15 inclusive Evidence may include values or indication on a diagram	AO1.1a	M1		$t_2 = 14.25$
	Deduces t_2 value correctly	AO2.2a	A1		
Total 5 marks					

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Finds correct acceleration	AO1.1b	B1	0.5 m s ⁻²
(b)	Identifies $5T$ as the distance travelled after the first 15 seconds	AO3.4	B1	Distance at constant speed = $5T$ Distance in first 15 secs =
	Uses the information given to form an equation to find T (award mark for either trapezium expression separate, totalled or implied)	AO3.1b	M1	$\frac{1}{2} \times (3 + 8) \times 10 + \frac{1}{2} \times (8 + 5) \times 5$ $= 55 + 32.5 = 87.5$ $5T + 87.5 = 120$ $\text{So } T = 6.5$
	Correctly calculates the distance for the first 15 secs	AO1.1b	A1	
	Deduces the values of T	AO2.2a	A1	

from the mathematical models applied		
Total 5 marks		

Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a)	States correct expression for a	AO1.1b	B1	$a = \frac{V - U}{T}$
(b)	Rearranges to make T the subject of the formula	AO2.1	R1	$T = \frac{V - U}{a}$
	Uses given expression for S and attempts to eliminate T	AO2.1	R1	$S = \frac{1}{2}(U + V) \times \frac{V - U}{a}$ $2as = (U + V)(V - U)$
	Completes argument to reach required result AG Only award if they have a completely correct solution, which is clear, easy to follow and contains no slips	AO2.1	R1	$V^2 = U^2 + 2aS$ (AG)
Total 4 marks				

Q5.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Calculates two (or four) appropriate distances	AO3.1b	M1	$s_1 = \frac{1}{2} (6 + 10) \times 8 = 64 \text{ m}$
	Obtains correct distances	AO1.1b	A1	$s_2 = \frac{1}{2} \times 10 \times 2 = 10 \text{ m}$
	Obtains correct sum of 'their' distances	AO1.1b	A1F	$s_1 + s_2 = 64 + 10 = 74 \text{ m}$ OR $S_1 = 6 \times 8 = 48 \text{ m}$ $S_2 = \frac{1}{2} \times 4 \times 8 = 16 \text{ m}$ $S_3 = \frac{1}{2} \times 4 \times 2 = 4 \text{ m}$

			$S_4 = \frac{1}{2} \times 6 \times 2 = 6 \text{ m}$ $S_1 + S_2 + S_3 + S_4 = 74 \text{ m}$
(b)	Finds difference of 'their' distances from part (a)	AO2.2a	B1F $64 - 10 = 54 \text{ m}$
(c)	Calculates magnitude of acceleration	AO3.1b	M1 $a_{\max} = \frac{8}{4} = 2$
	Obtains correct resultant force	AO1.1b	A1 $F_{\max} = 800 \times 2 = 1600 \text{ N}$
(d)	Explains that abrupt changes and straight lines in the graph are unlikely in reality	AO3.5b	E1 Change of velocity is unlikely to result in abrupt changes I would expect to see curves on the graph
Total 7 marks			

Q6.

(a) $s = \frac{1}{2} (5 + 4) \times 6 + \frac{1}{2} (4 + 7) \times 8 + 7 \times 7$

M1: Method based on three (or four or more!) areas / distances or equivalent added together.

A1: Correct calculation or value for one area / distance for one time period (eg 0 to 6 seconds).

M1A1

$= 27 + 44 + 49$

A1: Correct calculation or value for area / distance for another time period.

A1

$= 120 \text{ m}$

A1: Correct final distance.

A1

For example $24 + 44 + 49 = 117$ scores M1A1A1A0.

4

(b) Average Speed = $\frac{120}{21} = 5.71 \text{ m s}^{-1}$

M1: Their answer to part (a) divided by 21.

M1

A1F: Correct average speed.

Accept $5\frac{5}{7}$ or $\frac{40}{7}$.

A1F

2

[6]

Q7.

(a) $s_1 = \frac{1}{2} \times 5 \times 28 = 70 \text{ m}$

M1: For $\frac{1}{2} \times 5 \times 28$ or equivalent.
A1: Correct distance

M1A1

2

(b) $s = 70 + \frac{1}{2} \times 5 \times 22$

B1: For $\pm \frac{1}{2} \times 5 \times 22$ or equivalent.
M1: For adding the distances.

B1M1

$= 70 + 55$

$= 125 \text{ m}$

A1F: Correct distance. Follow through their answer from part (a) only.

A1F

3

(c) Average speed = $\frac{125}{50} = 2.5 \text{ ms}^{-1}$

M1: For their answer to (b) divided by 50.

M1

A1F: Correct average speed. Follow through answers from part (b).

A1F

2

(d) Displacement from $O = 70 - 55 = 15 \text{ m}$

B1: Correct displacement.

B1

1

(e) Average velocity = $\frac{15}{50} = 0.3 \text{ ms}^{-1}$

M1: For their answer to (d) divided by 50, provided they

have subtracted in (d).

M1

A1F: Correct average velocity. Follow through answers from part (d).

Award no marks if the final answer is 0.

A1F

2

(f) $a = \frac{5}{18} = 0.278 \text{ ms}^{-2}$

B1: Correct acceleration. Accept $\frac{5}{18}$ or equivalent fraction or 0.277 or AWRT 0.278.
Condone 0.28.

B1

1

[11]

Q8.

(a) $s = \frac{1}{2} \times 10 \times 4 + 10 \times 4 + \frac{1}{2} \times (4+7) \times 10 + \frac{1}{2} \times 7 \times 10$
 $(= 20 + 40 + 55 + 35)$

M1: Any one term correct.

M1: A second term correct.

A1: Correct expression for total distance.

M1M1A1

$= 150 \text{ m}$

A1: Total distance correct.

A1

OR

$s = \frac{1}{2} \times (10 + 20) \times 4 + \frac{1}{2} \times (4 + 7) \times 10 + \frac{1}{2} \times 7 \times 10$
 $(= 60 + 55 + 35)$
 $= 150 \text{ m}$

(M1M1A1)
(A1)

OR

$$\begin{aligned}
 s &= \frac{1}{2} \times 10 \times 4 + 10 \times 4 + 10 \times 4 + \frac{1}{2} \times 10 \times 3 + \frac{1}{2} \times 7 \times 10 \\
 &= 20 + 40 + 40 + 15 + 35 \\
 &= 150 \text{ m}
 \end{aligned}$$

(M1M1A1)
(A1)

4

(b) Average Speed = $\frac{150}{40} = 3.75 \text{ m s}^{-1}$
M1: Their total distance divided by 40.

M1

A1F: Correct average speed based on their distance from part (a). Must be correct to three or more significant figures.

A1F

2

(c) $a = \frac{4}{10} = 0.4 \text{ m s}^{-2}$
M1: Any division involving the numbers 10 and 4.

M1

A1: Correct acceleration. CAO

A1

Note on use of constant acceleration equations:
 award M1 for correct equation with correct values
 and A1 for correct final answer.

2

(d) $F = 200000 \times 0.4 = 80000 \text{ N}$

M1: Multiplication of, 2×10^n , for any integer n , by candidate's acceleration from part (c).

A1F: Correct force based on their answer to part (c) multiplied by 200000.

M1A1F

*Note: use of $a = 2.5$ gives 500000 N
 Accept 80 kN*

2

[10]

Q9.

(a) $s = 32 \times 12.5 = 400 \text{ m}$

B1: Correct distance.

B1

1

(b) $1600 = \frac{1}{2} (32 + 18)t$

M1: Seeing 2000 – candidate's answer to part (a) calculated

dM1: Use of constant acceleration equation(s) to find t , with $u = 32$ and $v = 18$

M1dM1

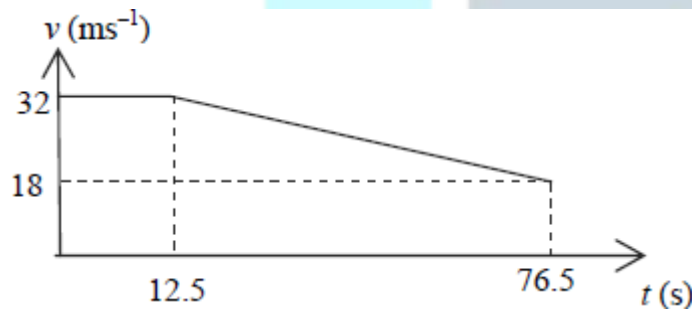
$$t = \frac{1600}{25} = 64 \text{ seconds}$$

A1: Correct time. Accept only 64

A1

3

(c)



B1: Shape of the graph.

B1

B1: Correct velocities (ie 18 and 32) on vertical axis.

B1

B1F: Correct times (ie 12.5 and 76.5) on the horizontal axis.

B1F

(Follow through incorrect answers to part (b)).

Award marks for graph if seen in earlier parts.

3

(d) Average Speed = $\frac{2000}{12.5 + 64} = 26.1 \text{ ms}^{-1}$

M1: Use of 2000 over candidate's total time (not 64 or 12.5).

M1

A1F: Correct speed. AWRT 26.1.

FT candidate's answer to part (b) or (c).

A1F

2

Q10.

- (a) 30 seconds

B1: Correct statement of time.

B1

1

- (b) $s_1 = \frac{1}{2} \times 40 \times 20 = 400 \text{ m}$

M1: A method for calculating the first distance. Must see 40 and $\frac{1}{2}$.

M1

A1: Correct distance.

A1

OR

$$s_1 = \frac{1}{2} \times (20 + 0) \times 40 = 400 \text{ m}$$

(M1)(A1)

OR

$$a = -\frac{20}{40} = -\frac{1}{2}$$

$$0^2 = 20^2 + 2\left(-\frac{1}{2}\right)s$$

$$s = 20^2 = 400 \text{ m}$$

Note on third method: Must see $-\frac{1}{2}$ or $-\frac{20}{40}$ plus attempt to find distance for M1.

(M1)(A1)

2

- (c) $s_2 = \frac{1}{2} \times 50 \times 20 = 500 \text{ m}$

M1: Method for finding the second distance and calculating the total distance.

M1

OR

$$s_2 = \frac{1}{2} \times (0 + 20) \times 50 = 500 \text{ m}$$

(M1)

OR

$$a = \frac{20}{50} = \frac{2}{5}$$

$$20^2 = 0^2 + 2\left(\frac{2}{5}\right)s$$

$$s = 20^2 \times \frac{5}{4} = 500 \text{ m}$$

Note on third method: Must see $\frac{2}{5}$ or $\frac{20}{50}$ plus attempt to find distance.

(M1)

$$\text{Total} = 400 + 500 = 900 \text{ m}$$

A1F: Correct total distance. Award the follow through mark for correct addition of 500 and their answer to (b).

A1F

2

(d) $v_{\text{AVERAGE}} = \frac{900}{120} = 7.5 \text{ ms}^{-1}$

M1: Their total distance divided by 120

M1

A1F: Correct average speed based on their answer to (c).

A1F

2

(e) $120 \times 20 - 900 = 1500 \text{ m}$

M1: Multiplication of 20 and 120 to find distance.

Note: Award M1 if 2400 seen in this part.

A1F: Correct difference based on their answer to (c) provided final answer is positive.

M1A1F

2

[9]

Q11.

- (a) $t = 0, t = 30, t = 50$ seconds

B1: Any one correct time

B1

*B1: The other two correct times
Deduct one mark for each extra time if more than three times are given.
(eg 0, 15, 30, 50 scores B1 B0)
(eg 0, 15, 30, 40, 50 scores B0 B0)
Condone 49 or 48 instead of 50*

B1

2

- (b) $s_1 = \frac{1}{2} \times 30 \times 5 = 75 \text{ m}$ **AG**

M1: Finding distance by calculation of area. (Must see use of 0.5 or $\frac{1}{2}$)

M1

*A1: Correct answer from correct working.
(If candidates use two constant acceleration equations, both must be seen for the M1 mark.)*

A1

2

- (c) $s_2 = \frac{1}{2} \times 4 \times 20 = 40 \text{ m}$

M1: Finding distance using area of the second triangle.

M1

*A1: Correct distance (ignore any negative signs).
(If candidates use two constant acceleration equations, both must be seen for the M1 mark.)
Accept 38/36 from use of 49/48 instead of 0*

A1

$$s = 75 + 40 = 115 \text{ m}$$

M1: Addition of the 75 metres and their distance. (75 – 40 = 35 OE scores M0)

M1

*A1F: Correct result using their value for second area.
eg Accept 113/111 from use of 49/48 instead of 50*

A1F

4

(d) $s = 75 - 40 = 35 \text{ m}$

M1: Difference between 75 and their value for the second distance.

(Allow their istance – 75)

(75 – (– 40) = 115 OE scores M0)

M1

A1F: Correct result using their value for second area.

(eg 40 – 75 = – 35 M1A0)

eg Accept 37/39 from use of 49/48 instead of 50

A1F

2

[10]

Q12.

(a) $s = \frac{1}{2}(3 + 10) \times 3$

Finding distance by summing 3 areas or using formula for the area of a trapezium

M1

Correct equation/3 correct expressions for the areas

A1

$= 19.5 \text{ m}$

Correct total distance

A1

3

(b) $a = \frac{3}{4} = 0.75 \text{ m s}^{-2}$

Correct acceleration as a decimal or as a fraction

B1

1

(c) $T - 400g = 400 \times 0.75$

Three term equation of motion containing

T , 400g and 400×0.75 or equivalent

M1

Correct equation

A1F

$$T = 3920 + 300 = 4220 \text{ N}$$

Correct tension

A1F

Only ft from $\alpha = \frac{4}{3}$

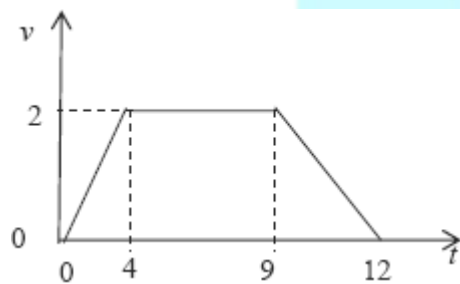
(ft 4453 N or 4450 N from $\alpha = \frac{4}{3}$ scores M1A1A1)

3

[7]

Q13.

(a)



Starts and finishes at rest

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B1

Correct shape

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B1

Correct values on t -axis

B1

Correct values on v -axis

B1

Condone omission of the origin

4

(b) $s = \frac{1}{2} (5 + 12) \times 2$

or $s = \frac{1}{2} \times 2 \times 4 + 5 \times 2 + \frac{1}{2} \times 2 \times 3 = 17$

Use of the area under the graph (or equivalent)

to find s

M1

$$= 17$$

Correct distance

SC When 21 used instead of 12 allow full marks for $s = 26$

A1

2

(c) $\max a = \frac{2}{4} = 0.5$

Maximum acceleration

B1

$$300 \times 0.5 = T - 300 \times 9.8$$

Three term equation of motion using their a

M1

Correct equation using $a = 0.5$

A1

$$T = 2940 + 150 = 3090$$

Correct tension

A1

4

[10]

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Q14.

(a) (i) $a = \frac{18 - 12}{10}$
 $\pm$

M1

$$a = 0.6 \text{ ms}^{-2}$$

A1

2

(ii) $s_1 = \frac{1}{2} (12 + 18) \times 10$

M1

$$s_1 = 150 \text{ metres}$$

A1

2

(b) $s_2 = 20 \times 12 = 240 \text{ metres}$

B1

1

(c) $s_3 = \frac{1}{2} (12 + 18) \times 20$

M1

$$= 300$$

PI

A1

$$\text{average speed} = \frac{150 + 240 + 300}{50}$$

M1

$$= 13.8 \text{ ms}^{-1}$$

ft one slip

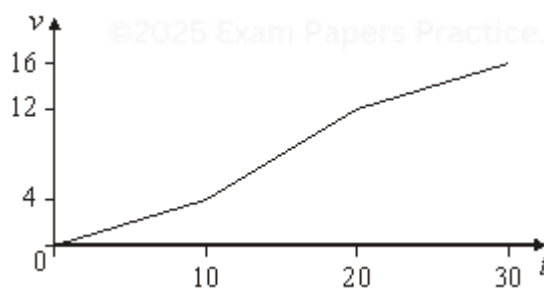
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4

[9]

Q15.

(a) (i)



3 straight lines

B1

correct end points

B1

sensible scales ÷ labelled v/t

B1

3

$$(ii) \quad s = \frac{1}{2} \times 10 \times 4 + \frac{1}{2} \times (4 + 12) \times 10 + \frac{1}{2} (12 + 16) \times 10$$

M1

Or
 area attempt (M1)
 full method (m1)
 equation correct (A1)
 ft one slip

m1

Or
 equation attempted (M1)
 full method (m1)
 all correct (A1)
 ft one slip

A1

$$s = 240 \text{ metres}$$

A1ft

4

$$(iii) \quad \text{Average speed} = \frac{240}{30}$$

M1

$$= 0.8 \text{ ms}^{-1}$$

cao

A1

2

(iv) Greatest acceleration = 2nd stage

$$= \frac{12 - 4}{10}$$

M1

$$= 0.8 \text{ ms}^{-2}$$

CAO

A1

2

(b) (i) Less

B1

area below curve < area below line/velocity lower
no additional incorrect statements

B1
2

(ii) Change in velocity more gradual oe

B1
1

[14]

Q16.

(a) $v = u + at$

$0 = 10 + (-0.8) \times t$

Full method with u, v used correctly, Accept

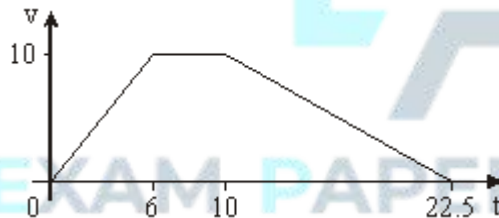
M1

$t = 12.5 \text{ sec}$

CAO, (correct substitution and answer)

A1
2

(b)



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each line, straight and correct end points

B1 B1 B1

*SC first error in times lose B1, subsequent
repeated errors, no further penalty*

*SC B1 for 3 lines giving correct shape but no values shown
axes labelled, v, t*

B1
4

(c) $\text{distance} = \frac{1}{2} \times 10 \times (4 + 22.5)$

full correct method (including addition of distances)

M1

correct subs, ft graph if final $t = 12.5$

A1

= 132.5 metres

ft one slip AWRT 133

A1F

3

[9]

Q17.

(a) $v = u + at$

$0 = 10 + (-0.8) \times t$

Full method with u, v used correctly

Accept ± 0.8

M1

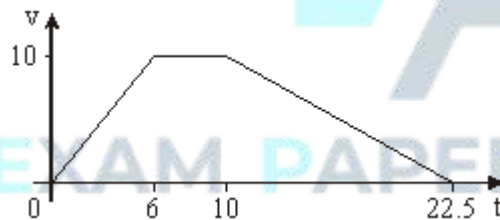
$t = 12.5 \text{ sec}$

CAO (correct subs and answer)

A1

2

(b)



each line, straight and correct end points

B1 B1 B1

SC: B1 for 3 lines giving correct shape but no values shown

SC: first error in labelling times loses B1, repeated errors no further penalty
axes labelled v, t

B1

4

(c) $\text{distance} = \frac{1}{2} \times 10 \times (4 + 22.5)$

Full correct method

M1

Correct subs, FT graph if final $t = 12.5$

A1F

= 132.5 metres

FT one slip, AWRT 133

A1F

3

- (d) Acceleration unlikely to:
change so abruptly or be constant
or velocity unlikely to be constant

B1

1

[10]

Q18.

(a) $s = \frac{1}{2} \times 15 \times 20 = 150$

M1: Finding length of first stage

M1: Finding length of second stage

A1: Both distances correct

M1

$$s = \frac{1}{2} \times 15 \times 80 = 600$$

M1A1

$$s = 600 + 150 = 750 \text{ m}$$

A1: Correct total distance

A1

4

(b) (i) $t = \frac{750}{15} = 50 \text{ s}$

B1: Correct time or their distance
correctly divided by 15

B1ft

1

(ii) Delay = $120 - 50 = 70 \text{ s}$

B1: Correct time of their previous time
correctly subtracted from 120 to give a
positive answer

B1ft

1

(c) $a = \frac{15}{80} = \frac{3}{16} = 0.1875 \text{ ms}^{-2}$

M1: Finding acceleration

A1: Correct acceleration

M1A1

$$F = 500000 \times 0.1875 = 93800 \text{ N (to 3sf)}$$

M1: Use of $F = ma$

A1: Correct force

M1A1

4

[10]

Q19.

(a) $20 = \frac{1}{2} (3 + V)4$

Full method, ($a = 1$)

M1

$$V = 7$$

A1

2

(b) (i) $S = 2V (=14)$

Alternative: Time = $T - 4$

M1

$$35 - S = \frac{1}{2} \times t \times V$$

Full method $35 = \frac{1}{2} (T - 4 + 2) \times V$

M1

$$21 = \frac{7}{2} t_2$$

Correct subs $35 = \frac{1}{2} (T - 2) 7$

A1

$$t = 6 \quad t = 12$$

$$\text{ft one slip} \quad T = 12$$

A1F

4

$$(ii) \quad f = \pm \frac{7}{6} = 1.17 \quad (1.1666)$$

$$\text{Accept } \pm \text{ ft time}$$

M1A1F

2

$$(c) \quad \text{Average speed} = \frac{\text{Total distance}}{\text{time}} = \frac{55}{12}$$

$$\text{Attempt at full area for distance}$$

M1

$$= 4 \frac{7}{12} \quad (4.5833)$$

$$\text{ft time}$$

A1F

2

[10]

Q20.

$$(a) \quad (i) \quad V = 0 + 3 \times 0.8$$

M1

$$V = 2.4$$

$$\text{AG}$$

A1

2

$$(ii) \quad S = \frac{1}{2} (5 + 8) \times 2.4$$

$$\text{Method for total area}$$

M1

A1

$$S = 15.6\text{m}$$

$$\text{ft slip}$$

A1F

(b) $\frac{1}{2} \times T \times 2.4 = 4.8$

Full method for time

M1

$T = 4$

A1

$$\text{Average speed} = \frac{15.6 + 4.8}{8 + 4}$$
Total distance / total time

M1

$= 1.7 \text{ ms}^{-1}$

ft one error in S or in T

A1F

4

[9]

Q21.

(a) (i) $T = 0.6a$

Either equation (M1 A0 for use of $0.1ga$)

M1

$0.1g - T = 0.1a$

SC who;e strng method:

A1

$0.1g = 0.7a$ M1A1 (total mass used)

A1

$0.1g = 0.7a$

$a = 1.4$ A1, max3/5)

m1

$a = 1.4 \text{ ms}^{-2}$

A1

5

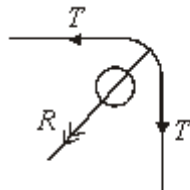
(ii) $T = 0.6 \times 1.4 = 0.84 \text{ N}$

Dependent on M1 gained in (a),
 Or, s.c. can gain M1 (from (a)) A1 here if
 equations involving T not found in (a)
 Max M1 A1

A1

1

(iii)



$$R = 2T \cos 45^\circ$$

M1

recognising 2 tensions involved

B1

M1: attempt at Pythagoras or at a component of T

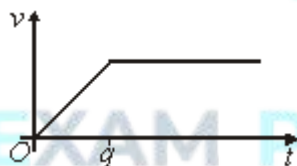
$$= 1.19 \text{ N}$$

A1: f.t. tension

A1F

3

(b)



1st line sloping and through O

B1

2nd line horizontal

B1

label at $t = q$

B1

3

[12]

Q22.

- (a) $a = \text{change in velocity} / \text{time or}$
used

M1

$$v = u + at$$

$$a = 2 \text{ m s}^{-2}$$

A1

2

(b) distance = $\frac{1}{2} \times 5 \times 6$
Full method

M1A1F

15 metres

A1F

3

- (c) acceleration not likely to be constant initial acceleration
 unrealistic change in velocity at $t = 3$ will be smoother

B1

1

[6]

Q23.

(a) $v = 0 + 0.8 \times 3$
Full method for M1

M1

$$v = 2.4$$


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A1

2

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(b)



1st line

2nd line

labels: 3, 7, 2.4

(2.4 may be implied by position between 2 and 3)

B1B1B1

3

(c) $s_1 = \frac{1}{2} \times 3 \times 2.4$ **or** $s_1 = 0 + \frac{1}{2} \times 0.8 \times 3^2$

M1

$$s_2 = 9.6$$

B1

$$s = 13.2$$

A1F

$$\text{average speed} = \frac{13.2}{7}$$

$$= 1.89 \text{ ms}^{-1}$$

A1F

5

Alternative

$$s = \frac{1}{2} (7 + 4) \times 2.4$$

$$= 13.2$$

M1A1

A1F

*Allow candidate's total dist. Found in (c)
ft total distance found in (c)*

[10]

Q24.

(a) (i) $a = \frac{(5U - 3U)}{(5 - 2)}$

full method for a Papers Practice. All rights reserved.

M1

$$a = \frac{2U}{3}$$

Accept 0.67U

A1

2

(ii) $s = \frac{1}{2} (5U + 3U) \times 3$

full method for s or attempt at area of whole shape

M1

$$s = 12U$$

ft incorrect a used

A1F

2

(b) $20 = \frac{1}{2}(5U + 0) \times 4$

*full method, use of equation or area of triangle
(a = -1.25 U if used)*

M1A1

$$U = 2$$

ft slip in line above e.g. miscopy

A1F

3

Alternative: average speed = 5

M1

equation for U

A1

$$U = 2$$

A1

[7]

Q25.

(a) *A:* $0.21g - T = 0.21a$
'whole string' method 3/5

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B: $T - 0.14g = 0.14a$

M1A1 for equation, A1 for answer

A1

$$0.07g = 0.35a$$

M1

$$a = 1.96 \text{ ms}^{-2}$$

CAO

A1

5

(b) $v = 0 + 1.96 \times 2$

M1

$$v = 3.92 \text{ ms}^{-1}$$

A1

2

(c) $0 = 3.92 - 9.8t$

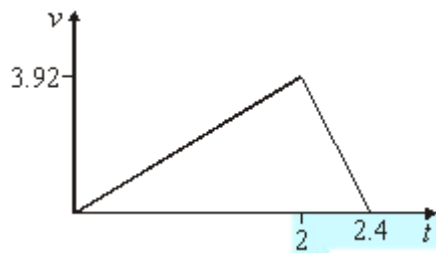
M1

$$t = 0.4 \text{ sec}$$

A1F

2

(d)



Two straight lines

Appropriate gradients

Labelling

B1

B1

B1

3

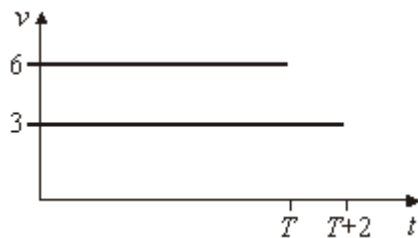
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[12]

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Q26.

(a)



horizontal line below other

B1

(0, 3 to (T + 2, 3)

B1

2

(b) (i) $x + 20$

B1
1

(ii) $x + 20 = 6T$

B1
1

(iii) $3(T + 2) = x$

M1A1
2

(iv) $6T - 3T - 6 = 20$

M1

$$3T = 26 \quad T = \frac{26}{3}$$

ft if T positive

A1F

$$\frac{26}{3} \times 6 = x + 20$$

M1

$$x = 32$$

ft if x positive

A1F

4

[10]

Q27.

(a) $s = \frac{1}{2} (40 + 60) \times 4$

Equation based on area of trapezium of rectangle plus two triangles

M1

Correct equation

A1

$= 200\text{m}$

Correct final distance

A1
3

(b) $a = \frac{4}{5} = 0.8 \text{ ms}^{-2}$

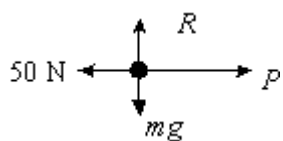
Attempting to find acceleration

M1

Correct acceleration

A1
2

(c) (i)



Correct force diagram

B1
1

(ii) $P - 50 = 65 \times 0.8$

Three term equation of motion

M1

Correct equation

A1

$P = 102 \text{ N}$

Correct value

A1
3

[9]